

Joint Call on Oceans: Climate Change Mitigation and Adaptation

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The deadline for proposal submission is

Tuesday, 22nd July 2025

10:00 (morning) Europe CEST / 17:00 (afternoon) Japan JST

Please make sure that the number of pages you prepare
for each section corresponds to the maximum number indicated in brackets.

Use Arial font in size 11.

Proposals exceeding the given limits will not be accepted.

Project title: Understanding Ocean Heat Content and Its Role in Driving Marine Heatwaves and Ecosystem Responses

Project acronym: UHEAT

Keywords: Ocean Heat Content, Marine Heatwaves, Meridional Heat transport, AMOC, Fish Habitats

Subtopic(s): Subtopic 2

External reviewers: Antonietta Capotondi (NOAA), Matthew D. Palmer (UK Met Office), Emanuele Di Lorenzo (Brown University).

Name and institution of project leader for Japan: Hiroshi Kuroda, Hokkaido University (HKU), Japan.

Name and institution of project leader for Europe: Francisco M. Calafat, University of the Balearic Islands (UIB), Spain.

(Underline the overall project leader from the 2 above)

Names and institutions of other project partners: Eleanor Frajka-Williams, University of Hamburg (UHH), Germany.

1. General information

1.1. Short abstract of the project

Please describe briefly the rationale and overall aim of the project and its relevance regarding the aim of the Joint Call (max. 300 words and up to 1 page when including graphs).

About 90% of the excess heat trapped by the Earth's climate system due to global warming has been absorbed by the ocean, leading to a significant increase in global ocean heat content (OHC). This heat is not stored locally but is instead redistributed both across the globe and into the ocean interior by ocean circulation, resulting in a highly uneven pattern of warming (Fig. 1). Importantly, the newly stored heat can alter ocean circulation itself, leading to anomalous ocean heat transport (OHT) and, in turn, disrupting the ocean's key role in regulating the regional climate and contributing to strong regional contrasts in ocean warming.

Furthermore, increased OHC makes waters more susceptible to marine heatwaves (MHWs), with significant impacts on marine ecosystems, including commercially important fish species.

The proposed project, UHEAT, aims to improve our understanding of regional OHC changes and their role in preconditioning MHWs in key regions of rapid ocean warming, and leverage this understanding to advance prediction of MHWs and their impact on key fish species. We will generate new datasets of OHC, OHT, and ocean circulation in the Atlantic and Pacific Oceans using statistically rigorous Bayesian methods that combine hydrographic and satellite observations. These data will enable new insights into the processes governing OHC changes, and provide a foundation for developing an artificial intelligence (AI) system to predict future MHW probabilities based on OHC, OHT and sea surface temperature (SST). In parallel, we will also assess future changes in OHC and MHW likelihood using climate model projections, applying emergent constraints techniques.

Finally, we will link predicted and projected MHWs to potential changes in the distribution of key commercially important fish species, using species-specific temperature ranges. This will provide actionable insights for fisheries management and climate adaptation, enhancing the societal relevance and impact of the research.

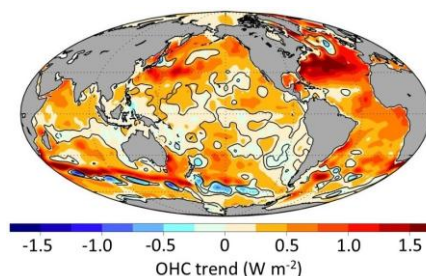


Figure 1. Upper-ocean heat content trends (0-700 m) for 1968–2019. (Johnson & Lyman, 2020).

1.2. Main objectives of the project

Please describe the overall aim and specific objectives in detail, state of the art, giving when possible quantitative data. Describe also the expected results of the project (max. 300 words and up to 1 page when including graphs).

UHEAT will advance understanding of regional OHC changes and their connection to MHWs, and exploit this understanding to develop AI-based models for the prediction of MHWs and their impacts on key fish species. Recent advances in ocean observing and climate modelling have enabled significant progress towards explaining OHC changes. Changes in global average OHC since 1960 are

now relatively well constrained, and there is reasonable confidence in the larger-scale patterns of OHC trends since 2005 thanks to the Argo program, such as enhanced warming of the North Atlantic and western North Pacific. However, important uncertainties remain in regional patterns and variability even in the Argo era, due to sparse observations and methodological inconsistencies. In addition, the links between OHC changes, OHT and ocean circulation are not yet fully understood. UHEAT will leverage advanced data merging techniques and state-of-the-art climate model outputs to quantify and explain regional OHC changes and their role as a driver of MHWs. UHEAT will develop AI-based tools to generate predictions of MHW likelihood conditional on OHC, OHT and SST, and will translate these predictions into ecosystem-relevant information, supporting fisheries management and climate adaptation strategies.

UHEAT has five main scientific objectives:

1. Generate observation-based records of OHC, OHT and ocean circulation (1993 to present) across the whole Atlantic and Pacific Oceans using a Bayesian multi-observational approach.
2. Determine the key processes of recent OHC changes, including their links to ocean circulation, and assess the role of such changes as a preconditioning factor to MHWs.
3. Assess future changes in OHC and long-term shifts in MHW probabilities.
4. Develop an AI-based system for seasonal-to-multiyear forecast of MHW conditional on OHC, OHT and SST fields.
5. Predict the influence of MHWs on the distribution and migration routes of commercially important fishery resources in Japan and Europe.

1.3. The project description

Please explain the project in detail (max. 1200 words and up to 4 pages when including graphs).

1.3.1. Background

Global warming results from the increasing concentration of greenhouse gases, which traps extra heat in the Earth's climate system. About 90% of the excess heat trapped since the Industrial Revolution has been absorbed by the ocean (Von Schuckmann et al., 2023), leading to a significant increase in global average OHC. While this heat enters the ocean through the surface, it is not stored locally. Instead, ocean circulation redistributes it both laterally across the oceans and into the ocean interior, strongly influencing the patterns of ocean warming. Importantly, the newly stored heat does not act as a passive tracer but can instead modify the very currents that transport it (Marshall et al., 2015), resulting in anomalous OHT. The changes in OHT induced by the extra heat are central, in many regions, to setting the patterns of ocean heat uptake and warming. For example, the so-called warming hole in the subpolar gyre of the North Atlantic can be largely explained by reduced meridional OHT into the subpolar region due to a weakening of the Atlantic meridional overturning circulation (AMOC) (Marshall et al., 2015; Li and Liu, 2025). In the western North Pacific, the OHT comes largely from wind-driven gyre circulation within the thermocline rather than the meridional overturning circulation (Ferrari and Ferreira, 2011; Minobe, 2025).

Over the past several decades, the subtropical North Atlantic, western North Pacific, and Mediterranean Sea have experienced rapid increases in OHC (Johnson and Lyman, 2020), warming faster than the global average. In 2024, the OHC in all these regions was the highest on record (Cheng et al., 2025). In addition to making storms more severe (Kossin et al., 2020; Argüeso et al., 2024), ocean warming increases the likelihood of MHWs by predisposing waters to their occurrence and raising the baseline temperature on which they develop (Benthuisen et al., 2020; Capotondi et al., 2024). East of northern Japan, the record-breaking MHW observed during the 2020s has been linked to changes in the Kuroshio and Oyashio western boundary currents (Fig. 2) (Minobe, 2025), highlighting the key role of OHC and OHT in MHWs occurrence. MHWs can have profound impacts on marine ecosystems, including recurrent mass mortality events (Garrahou et al., 2022), habitat shifts in certain fish species (Caputi et al., 2016; Miyama et al., 2021), and widespread seagrass die-offs (Strydom et al., 2020). Explaining regional OHC changes and predicting MHWs is crucial to the development of adaptation strategies that can reduce the increasing vulnerability of marine ecosystems to the threats of ocean warming.

1.3.2. Current challenges

Extensive research efforts in recent decades have been devoted to quantifying past OHC changes (Von Schuckmann et al., 2023). Such efforts have enabled significant progress in our understanding of OHC changes, particularly on global and large spatial scales, including the 2023-2024 global boiling (Minobe et al. 2025). However, key knowledge gaps persist, such as questions about the spatial structure and magnitude of the OHC changes on smaller regional scales, the timescales of variability, the links between the OHC changes and ocean circulation, and the role of these oceanic factors as drivers of MHWs.

Estimates of regional OHC trends and variability derived from hydrographic data alone remain subject to large uncertainties, even in the Argo era (post-2005), because of the sparse and uneven distribution of observations in time and space. While ocean reanalyses offer improved spatiotemporal coverage, they carry their own biases and uncertainties, and crucially depend on assimilating the same in-situ data, facing many of the same limitations.

In the Atlantic, long trans-basin records of the AMOC and meridional OHT from direct ocean observations, which are essential for understanding the role of ocean circulation in OHC changes, are limited to two latitudes (Frajka-Williams et al., 2019): 26°N from the RAPID array (Johns et al., 2023), and 50°N-60°N from the OSNAP program (Lozier et al., 2019). For the North Pacific, there are several repeat observation lines, including one along 137°E by Japan, but coverage of continuous observations is also very limited. These observing gaps restrict our ability to assess the links between OHC and ocean circulation.

Assessing future changes in OHC is also challenging due to large uncertainties in climate model projections. Indeed, the IPCC (IPCC, 2021) has assessed regional and sub-basin scale projections of OHC change as having “low confidence” due to inconsistencies among climate models, particularly in boundary currents, marginal seas such as the Mediterranean Sea and the Japan Sea, and high-latitude regions.

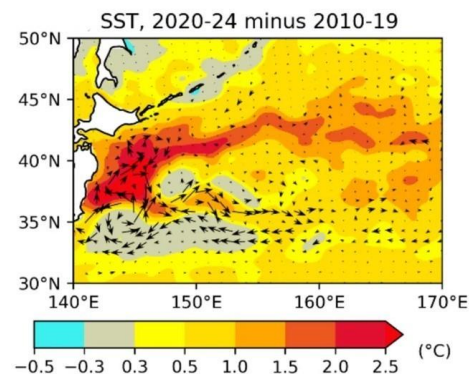


Figure 2. Western Pacific marine heatwave. Differences in SST, and 0-300 m ocean transport (vectors) between the 2020–2024 and 2010–2019 periods. (Minobe, 2025).

1.3.3. The UHEAT approach

UHEAT will address these challenges as outlined below.

Measuring OHC changes and OHT:

Recently, we developed a novel Bayesian hierarchical framework for estimating Atlantic meridional OHT that combines data from hydrography, satellite altimetry and satellite gravimetry (Fig. 3), exploiting cross-variable spatial information and rigorously accounting for data error structures (Calafat et al., 2025). This approach leads to significantly improved estimates of meridional OHT compared to those derived from hydrographic data alone. Our approach also produces spatiotemporal fields of OHC and OHT convergence as a by-product. Currently, it can only integrate gravimetry data from GRACE, limiting our estimates to the period starting in 2004. In UHEAT, we will extend our estimates back to 1993 by incorporating ocean mass data from additional sources, and will also produce estimates for the Pacific Ocean. We will also adjust it to allow for separation of upper and lower OHC changes. In parallel, we will develop a new method, also combining hydrography, altimetry and gravimetry, to estimate ocean circulation.

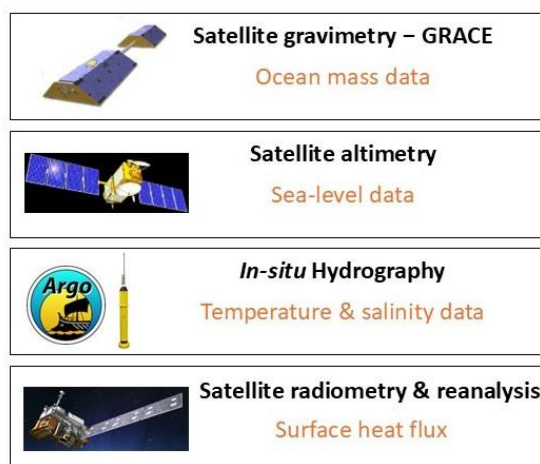


Figure 3. Observational data used in the Bayesian approach.

Constraining future changes in OHC and MHW probabilities: Assessing future changes is challenging due to biases and large uncertainties in climate model projections, arising from differences in model physics, resolution, and representation of key processes. UHEAT will use emergent constraints (Hall et al., 2019) to address this challenge. This approach exploits robust and physically interpretable relationships between present-day observable changes and modelled future changes across a model ensemble, to refine projections and effectively reduce uncertainty in climate projections. It also helps prioritize observations by identifying which measurements most effectively constrain future change, and has already improved projections of climate sensitivity and precipitation (Cox et al., 2018; Shiogama et al., 2022), among others.

MHW prediction: Deep learning is rapidly emerging as a powerful tool for predictive modelling across many fields, including climate science. While several recent studies have applied deep-learning methods to predict MHWs (Parasyris et al., 2025; Davenport et al., 2024), they have relied solely on SST as a predictor, despite growing evidence that OHC and OHT act as important preconditioning factors and retain predictive skill over longer lead times than atmospheric variables (Behrens et al., 2019; McAdam et al., 2023; Goes et al., 2024; Elzahaby et al., 2021). In UHEAT, we will develop a deep-learning prediction system for MHWs that incorporates OHC, OHT, and ocean circulation alongside SST to deliver more robust and skilful forecasts. Moreover, we will assess both surface and subsurface (0–50 m) MHWs, recognizing that subsurface warming more directly affects marine ecosystems, including fisheries.

Fisheries impact assessment: UHEAT will translate knowledge of OHC and MHW changes into ecosystem-relevant predictions of impacts on fish habitats and fisheries, supporting adaptation strategies for sustainable management in both Japan and Europe, thus directly delivering on the call's aim to improve understanding of ocean processes to enable effective climate change adaptation.

1.4. Scientific excellence of the project and the project partners

Please explain what makes up the excellence of the proposal and the consortium, and its innovative aspects (max. 600 words and up to 2 pages when including graphs).

UHEAT is an interdisciplinary project that will advance understanding of regional OHC changes and their role in driving MHWs, while delivering improved prediction tools and ecosystem-relevant information. Specific innovations include:

1. **Advanced Bayesian data integration of in-situ and satellite data** to produce improved, uncertainty-aware estimates of OHC, OHT and ocean circulation across the entire Atlantic and Pacific Oceans.
2. **Reducing uncertainty in future projections of OHC and MHW likelihood** through physically interpretable emergent constraints.
3. **Deep-learning prediction system that go beyond SST-only approaches** by incorporating OHC, OHT, and ocean circulation to deliver more mechanistically informed forecasts of MHW occurrence.
4. **Translation of physical predictions into actionable, ecosystem-relevant information** on fish habitat shifts to directly support adaptation strategies.

UHEAT brings together a highly complementary consortium of researchers with demonstrated excellence in physical oceanography, numerical climate simulation, marine ecosystem analysis, and statistical modelling.

The **Spanish team** (PI F. M. Calafat and Co-I M. Marcos) brings recognized expertise in ocean dynamics, sea-level physics, climate extremes, and advanced statistical modelling. They pioneered the development of Bayesian hierarchical models to estimate meridional OHT in the Atlantic through the H2020 EPOC project (101059547), delivering one of the first multi-observational, statistically rigorous estimates of Atlantic meridional OHT (Calafat et al., 2025). This provides a solid foundation for extending these methods to the Pacific Ocean and earlier periods, where comparable products are lacking. They have also led the development of other novel Bayesian frameworks for sea-level rise (Calafat et al., 2022a) and storm surge extremes (Calafat & Marcos, 2020; Calafat et al., 2022b) through the H2020 EuroSea project (862626) and the MICIU-funded CLIMEX project (CNS2023-145309). They also bring expertise in analysing MHW changes (Marcos et al., 2025) and developing AI-based forecasting systems through the MICIU-funded DETECT project.

The **Japanese team** (PI H. Kuroda, Co-I Y. Sasaki, and collaborator S. Minobe) bring recognized expertise in ocean dynamics in the North Pacific (Sasaki et al., 2013), including its relationship with OHC (Minobe, 2025), in big-data analysis involving outputs from climate models (Terada & Minobe, 2018; Abe & Minobe, 2023), and in MHWs and their impacts on marine ecosystems, including commercially important fish species (Kuroda & Yokouchi, 2017; Kuroda et al., 2020; Miyama et al., 2021; Minobe et al., 2025).

The **German team** (PI E. Frajka-Williams) contributes recognized expertise in large-scale ocean circulation and climate variability, particularly the AMOC and its links to heat transport and meridional connectivity. Their prior work includes diagnosing AMOC variability and its causes, using ocean observations as well as climate models and ocean reanalyses (Frajka-Williams, 2015; McCarthy et al., 2015; Frajka-Williams et al., 2019; Sanchez-Franks et al., 2021). Their contribution ensures that the physical interpretation of OHC and OHT patterns remains dynamically sound and tied to known modes of variability.

The scientific excellence of this consortium lies in their unique ability to integrate physical understanding, advanced statistical methods, and biological impact assessment into a cohesive, interdisciplinary framework. Furthermore, each partner institution is actively engaged in international ocean-climate science programs,

ensuring high visibility and dissemination of the project's findings through publications, open-access tools, and stakeholder networks.

1.5. Project coordination and management

Please describe how the project will be coordinated, and what will be the tasks of each partner (max. 600 words and up to 2 pages when including graphs).

UHEAT will be managed through a clear and robust coordination structure designed to ensure strong collaboration, effective communication, scientific integration, and timely delivery of outputs. The project will have two Project Leaders, as required by the call: Francisco M. Calafat (University of the Balearic Islands, UIB, Spain) serving as the Principal Project Leader with overall coordination responsibility, and Hiroshi Kuroda (Hokkaido University, HKU, Japan) as the Japanese Project Leader.

Principal Project Leader (F. M. Calafat): F. M. Calafat will act as the Principal Project Leader, providing overall scientific and administrative leadership for UHEAT. He will serve as the main contact point with the call secretariat, oversee progress across all work packages, and ensure that deliverables, reports, and financial documents are submitted on time and meet quality standards. He will chair regular project meetings, facilitate integration between partners, and manage risks and mitigation plans.

Japanese Project Leader (H. Kuroda): H. Kuroda will serve as the Project Leader on the Japanese side, coordinating all research activities carried out by the Japanese team and acting as the main point of contact with Japanese funding agencies. He will ensure that the Japanese contribution aligns with the overall project goals and timeline, and will work closely with the Principal Project Leader to ensure scientific and administrative coherence.

The German partner (University of Hamburg, UHH), led by PI Eleanor Frajka-Williams, will coordinate research activities on the German side and support the overall coordination of the project. UHH will actively participate in regular project meetings and joint planning, helping to maintain alignment of project activities with timelines and deliverables. They will contribute to collaborative decision-making and ensure effective communication among partners to support smooth and efficient project execution.

Overall Coordination Approach: Coordination will be maintained through regular virtual meetings every two months to review progress, discuss challenges, and plan next steps. Annual face-to-face meetings (alternating between Europe and Japan) will allow for in-depth scientific integration, strategy development, and stakeholder engagement. A shared online collaboration platform will be used for data exchange, document sharing, and collaborative work on publications and reports. All members of the team will contribute to results discussion and to preparation and writing of papers.

Each partner will also organize smaller internal meetings as needed to ensure alignment within their teams. UIB will take primary responsibility for project-wide administration, including timeline management and reporting. HKU and UHH will be responsible for managing reporting requirements and administrative obligations for the Japanese and German sides, respectively.

The detailed scientific roles and tasks of each partner, specifying their contributions to the project's research objectives and work packages, are provided in [Section 2.4](#).

Summary of the data management plan

The proposed work will be largely based on existing datasets (*in-situ* hydrographic data, satellite altimetry and gravimetry measurements, data from climate models and reanalyses, etc.) that are available from public archives. No new data will be collected, but we will produce several new data sets, including gridded estimates of OHC changes and OHT convergence, meridional OHT across multiple latitude lines, and

ocean circulation, among other data sets. These data sets will be deposited in a public repository under the Creative Commons Public license such as Zenodo where they will be assigned a permanent DOI which can then be used to cite the data. We will also develop substantial amounts of code to analyse the data and to implement the Bayesian hierarchical model. The code will also be deposited in Zenodo under a GNU General Public License.

2. Work plan

Please describe the work plan according to the four aspects below (max. 600 words per aspect and up to 2 pages when including graphs).

2.1. Research methodology

The project is organised in five inter-related work packages (WPs) designed to address the five scientific objectives (Fig. 4).

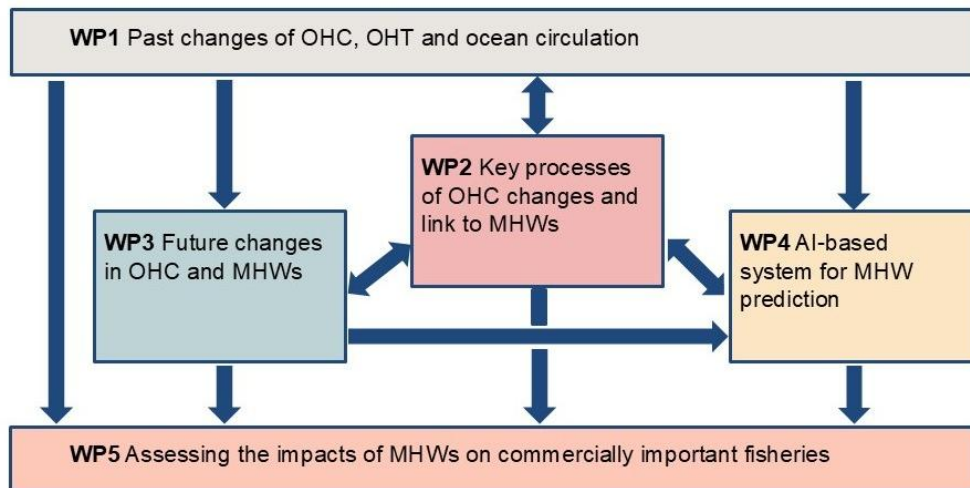


Figure 4. Project structure and WPs.

WP1: Past changes of OHC, OHT and ocean circulation

We will build on an existing Bayesian hierarchical framework developed for the Atlantic (Calafat et al., 2025) that integrates data from hydrography, altimetry, and gravimetry (GRACE) with surface heat flux data to generate gridded estimates of OHC changes and OHT convergence, and meridional OHT across multiple latitude lines. We will extend this framework to the Pacific Ocean, to earlier periods back to 1993 by incorporating additional ocean mass datasets beyond GRACE, and to enable separation of upper and deep OHC contributions. The Bayesian model also provides consistent fields of sea level and ocean mass which will be used to approximate the ocean circulation as a 2-layer fluid, providing barotropic and baroclinic estimates of ocean circulation (Cabanès et al., 2008).

WP2: Key processes of OHC changes and link to MHWs

Bringing together the observation-based data from WP1 and a suite of high-resolution models and reanalyses, WP2 will determine the processes of OHC changes since 1993. The core set of model simulations will come from the HighResMIP project (Haarsma et al., 2016), but these will be supplemented with state-of-the-art reanalyses (e.g., ECMWF ORAS5 – Zuo et al. 2019). We will compute regional heat budgets to determine whether observed OHC changes are driven by OHT or surface heat fluxes. An assessment will be made to determine the relationship of OHT with the AMOC and

the gyre circulations. We will also combine surface geostrophic velocities from altimetric sea level with our Bayesian estimates of upper OHT convergence to map horizontal heat transport pathways. We will evaluate the spatiotemporal coherence of the OHC and the role of local and remote oceanic forcing in setting this coherence. Finally, we will determine how OHC changes relate to the occurrence of MHWs.

WP3: Future changes in OHC and MHW probabilities

We will use the emergent constraints approach (Hall et al., 2019; Cox et al., 2018) to reduce uncertainty in projections of future OHC changes and MHW likelihood from climate models. By identifying robust relationships between present-day observed variability and simulated future change in model ensembles, we will constrain projections, leveraging outputs from CMIP6 (Eyring et al., 2016). Observed OHC changes from WP1 will provide essential constraints.

WP4: AI-based system for MHW prediction

We will design deep-learning models (e.g., CNNs and RNNs) to predict the probability of MHW occurrence at seasonal-to-multiyear lead times. Unlike existing SST-only approaches, our models will incorporate OHC, OHT, and circulation data to capture preconditioning processes. To address the limited samples of observations, we will apply transfer learning: pre-training on ensembles of climate simulations and fine-tuning using observational data from WP1 and reanalyses. The ensembles will come from the Decadal Climate Prediction Project (DCPP; Boer et al., 2016), providing a data sample on the order of thousands of years. DCP simulations will be calibrated through comparison to observations (various approaches will be explored). Predictions will cover both surface and subsurface (0–50 m) MHWs.

WP5: Assessing the impacts of MHWs on commercially important fisheries

We will translate physical predictions of SST and MHW occurrence into ecosystem-relevant information. For Japan, this includes mapping habitat suitability changes for key commercial fish species based on their thermal preferences. For Europe, we will co-develop a plan for extending assessments to the Atlantic/Mediterranean in collaboration with European fisheries experts. To support adaptation planning in both Japan and Europe, UHEAT will organize one dedicated workshop with fisheries stakeholders in each region, where we will present the project's results and gather feedback on needs and priorities.

2.2. Work plan (timetable / Gantt chart)

The UHEAT work plan is structured around five interconnected WPs, designed to advance the understanding and prediction of OHC changes, MHWs, and associated impacts on marine ecosystems. WP1 provides the foundational observational datasets through innovative Bayesian methods. WP2 builds upon these datasets to identify processes driving OHC variability and their links to MHWs. WP3 leverages emergent constraints to refine future projections of OHC and MHW probabilities. WP4 develops state-of-the-art deep-learning prediction systems for operational prediction of MHWs, informed by observational data and climate simulations. WP5 translates physical predictions into ecosystem-relevant assessments, focusing on impacts for commercially important fisheries. Below we outline specific tasks for each WP, followed by a Gantt chart summarizing their timing.

WP1: Past changes of OHC, OHT and ocean circulation

(Lead: UIB)

- **T1.1:** Collect and process observational data.
- **T1.2:** Extend Bayesian framework to Pacific, earlier periods (1993-present), and separate upper/deep OHC.
- **T1.3:** Develop multi-observational approach to estimate ocean circulation.
- **T1.4:** Generate estimates of gridded OHC, OHT, OHT convergence and ocean

circulation.

WP2: Key processes of OHC changes and link to MHWs

(Lead: UHH)

- **T2.1:** Assemble and preprocess outputs from HighResMIP simulations and ocean reanalyses.
- **T2.2:** Compute regional heat budgets to quantify roles of OHT vs. surface fluxes.
- **T2.3:** Assess links between OHC changes, OHT, AMOC, and gyre circulations, and map pathways of heat transport.
- **T2.4:** Analyse coherence of OHC and diagnose local vs. remote forcings.
- **T2.5:** Quantify links between OHC changes and MHW occurrence.

WP3: Future changes in OHC and MHW probabilities

(Lead: HKU)

- **T3.1:** Assemble and preprocess outputs from CMIP6 simulations.
- **T3.2:** Identify emergent constraint relationships linking observed and projected changes of OHC.
- **T3.3:** Apply emergent constraints to quantify future OHC changes and probabilities of MHW occurrence.

WP4: AI-based system for MHW prediction

(Lead: UIB)

- **T4.1:** Assemble DCPD ensembles and apply bias-correction/calibration relative to observations.
- **T4.2:** Design AI-based model (CNN/RNN) for MHW prediction integrating OHC, OHT, ocean circulation, and SST.
- **T4.3:** Pre-train AI-based model on DCPD ensembles.
- **T4.4:** Refine AI-based model using observational data through transfer learning.
- **T4.5:** Implement prediction tool for surface and subsurface MHW probabilities.

WP5: Assessing the impacts of MHWs on commercially important fisheries

(Lead: HKU)

- **T5.1:** Compile fisheries data and thermal preferences for key fish species (initially Japan-focused).
- **T5.2:** Analyse the impacts of recent MHWs on fisheries in the northwest Pacific using publicly available catch statistics and government reports.
- **T5.3:** Map habitat suitability changes based on seasonal-to-multiyear predictions and long-term projections of MHWs.
- **T5.4:** Co-develop a plan for extending assessments to the Atlantic/Mediterranean in collaboration with European fisheries experts.

A Gantt chart outlining the work plan and showing how long each task will take and when deliverables/milestones are expected to be completed is shown in Fig. 5. This plan will act as a reference to monitor the project progress between planned and actual activities and identify any aspects requiring attention to ensure goals are achieved within the stated time frame.

Computing and data storage

Training the AI-based prediction model will likely require substantial amounts of data, such as outputs from the DCPD simulations, and will be computationally intensive. The research group at UIB has access to several computing systems to support these demands, including 'Thor' (128 CPUs, 1.5 TB RAM, two NVIDIA RTX 6000 ADA GPUs), 'Harrier' (24 CPUs, 256 GB RAM, basic GPU), and 'Ada' (256 CPUs, 0.5 TB RAM). In addition, the group maintains a data storage server with a capacity of 180 TB.

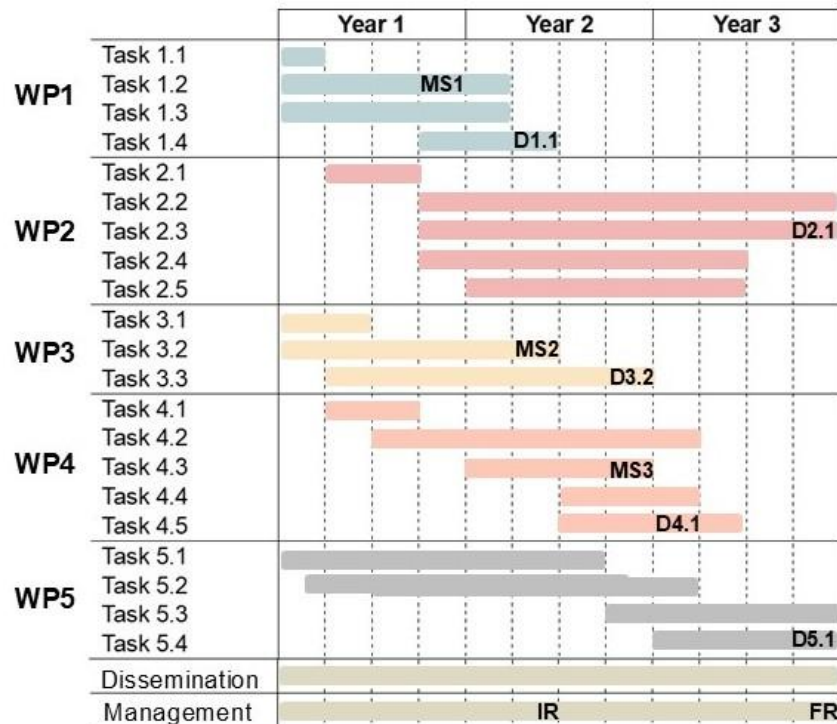


Figure 5. Gantt chart illustrating the schedule of tasks, deliverables, and milestones. IR: interim report, FR: final report.

2.3. Envisaged types of activities (including main milestones, deliverables and place in case of events)

2.3.1. Deliverables and Milestones

The milestones and deliverables of the project are outlined below along with their timing (see also Fig. 5).

WP1: Past changes of OHC, OHT and ocean circulation

- **Milestone (MS1):** Bayesian framework successfully extended to Pacific basin and earlier periods (Month 12).
- **Deliverables:**
 - **D1.1:** Observation-based estimates of gridded OHC, OHT convergence, OHT and ocean circulation (Month 18).

WP2: Key processes of OHC changes and link to MHWs

- **Deliverables:**
 - **D2.1:** Assessment of key processes of OHC changes, including pathways of heat transport, and link with MHWs (Month 36).

WP3: Future changes in OHC and MHW probabilities

- **Milestone (MS2):** Identification of initial emergent constraint relationships using observational and CMIP6 model data (Month 15).
- **Deliverables:**
 - **D3.1:** Assessment of future MHW probabilities under climate change scenarios using constrained projections (Month 24).

WP4: AI-based system for MHW prediction

- **Milestone (MS3):** Completion of AI model pre-training with DCPD ensembles (Month 24).
- **Deliverables:**
 - **D4.1:** Fully trained and validated AI prediction model integrating OHC, ocean circulation, and SST data (Month 24).

WP5: Assessing impacts of MHWs on commercially important fisheries

- **Deliverables:**
 - **D5.1:** Assessment of habitat suitability and fisheries impacts under predicted MHW conditions (Month 36).

In addition to the deliverables directly related to the research activities outlined above, the Principal Project Leader will also submit an interim report (IR) at month 18 and a final report (FR) at month 36 to the Program Secretariat. These reports will highlight the main results and outputs of the project, as well as any problems that have arisen and how they have been addressed.

2.3.2. Project meetings and workshops

We plan to hold a face-to-face kick-off meeting for the entire team at the start of the project, hosted by UIB in Spain. Two additional in-person Progress Meetings will take place at the end of Years 1 and 2 to review progress and ensure effective communication among all project partners. The first Progress Meeting will be hosted by HKU in Japan, and the second by UHH in Germany. In addition, we will organize one dedicated workshop with fisheries stakeholders in each region to present the project's results, demonstrate the forecasting tools, and gather feedback on needs and priorities. The first workshop will take place in Japan immediately after the first Progress Meeting, while the second will be held in Europe following the second Progress Meeting. A Final Project Meeting will also be held in person at UIB in Spain to discuss the project's results, conclusions, and scientific outcomes.

In addition, the team will participate in the two EIG CONCERT-Japan in-person workshops scheduled to take place in Europe. At least three team members, including representatives from both the European and Japanese partners, will attend each workshop.

2.4. Role of each partner

The UHEAT consortium is designed to ensure a well-coordinated, complementary partnership in which each team leads or contributes to specific tasks aligned with their expertise.

The **Spanish partner**, UIB, led by PI Francisco M. Calafat with Co-I Marta Marcos, will lead WP1 and WP4. In WP1, UIB will oversee collecting and processing observational data (T1.1), extending the Bayesian framework to cover the Pacific basin and earlier periods (T1.2), and generating gridded estimates of OHC, OHT, OHT convergence, and ocean circulation (T1.4), while contributing to the development of the multi-observational approach for ocean circulation led by UHH (T1.3). In WP2, UIB will lead the computation of regional heat budgets to quantify the roles of OHT versus surface fluxes (T2.2) and will perform the mapping of pathways of heat transport as part of the broader assessment of links between OHC changes, OHT, AMOC, and gyre circulations (T2.3). UIB will also lead T2.5, quantifying links between OHC changes and MHW occurrence, with contributions from HKU. In WP4, UIB will design, train, and refine the AI-based prediction system for MHW occurrence, integrating OHC, OHT, circulation, and SST data (T4.1 to T4.5). UIB will also coordinate overall project management, reporting, data-sharing, and dissemination, ensuring smooth collaboration across the consortium.

The **Japanese partner**, HKU, led by PI Hiroshi Kuroda with Co-I Yoshi Sasaki and collaborator Shoshiro Minobe, will lead WP3 and WP5. In WP3, HKU will be responsible for assembling and preprocessing CMIP6 simulation outputs (T3.1), identifying emergent constraint relationships linking observed and projected OHC changes (T3.2), and applying these constraints to quantify future OHC changes and MHW probabilities (T3.3). In WP5, HKU will compile fisheries data and thermal preferences for key fish species (T5.1), evaluate MHW impacts on stock recruitment (T5.2), map habitat suitability changes under seasonal-to-multiyear predictions and long-term projections of MHWs (T5.3), and develop a plan for extending these assessments to the Atlantic (T5.4). HKU will also contribute to tasks T2.2–T2.5, supporting the analysis of drivers of OHC changes and MHWs. Additionally, HKU will ensure that ecological and fisheries components are well integrated with physical predictions, delivering stakeholder-relevant outputs for Japan.

The **German partner**, UHH, led by PI Eleanor Frajka-Williams, will lead WP2 and play a key role in WP1. In WP1, UHH will lead T1.3, developing the multi-observational approach to estimate ocean circulation, and contribute to the generation of new data sets (T1.4). In WP2, UHH will coordinate the overall work package and lead the assembly and preprocessing of model and reanalysis outputs (T2.1), the assessment of links between OHC changes, OHT, AMOC, and gyre circulations (T2.3), the analysis of coherence of OHC and diagnosis of local versus remote forcings (T2.4), and will support UIB in delivering T2.2 and T2.5. UHH will also ensure that observational, modelling, and analytical approaches are effectively integrated, participating actively in coordination activities and project meetings to maintain alignment across all partners.

This clear distribution of roles and tasks will ensure that each partner's strengths are fully leveraged to achieve UHEAT's scientific objectives while fostering effective collaboration throughout the project.

3. Project budget

Type of costs	Partner	2026 (April~)	2027	2028	2029 (~Mar)	TOTAL
Research materials	HKU, Japan	9,677	10,484	9,677	2,419	32,257
	UIB, Spain	2,000	0	0	0	2,000
	UHH, Germany	0	0	0	0	0
Personnel costs	HKU, Japan	4,839	6,452	6,452	1,613	19,356
	UIB, Spain	26,000	51,000	26,000	0	103,000
	UHH, Germany	28,000	37,000	37,000	8,000	110,000
Travel*	HKU, Japan	7,258	9,677	9,677	2,419	29,031
	UIB, Spain	1,200	7,200	7,200	1,200	16,800
	UHH, Germany	1,200	3,000	2,000	1,200	7,400
Living expenses	HKU, Japan	1,452	1,935	1,935	484	5,806
	UIB, Spain	900	6,150	6,150	900	14,100
	UHH, Germany	800	1,200	600	1,000	3,600
Organisation of events	HKU, Japan	968	1,290	1,290	323	3,871
	UIB, Spain	700	0	0	1500	2,200
	UHH, Germany	0	0	1,500	0	1,500
Eligible indirect costs	HKU, Japan	7,258	8,952	8,710	2,177	27,097
	UIB, Spain	7,700	16,088	9,838	900	34,526
	UHH, Germany**	6,000	8,240	8,220	2,040	24,500
TOTAL		105,952	168,668	136,249	26,175	€437,044

*The team should budget for two EIG CONCERT-Japan in-person workshops in Europe (organized with the other funded teams in the same call and funding agencies). Each team is expected to have at least three members attending each workshop, including representation from both the European and Japanese sides. The Program Secretary will contact the Lead PI to share dates and details for the workshops.

**Indirect costs are not eligible for Germany, but a flat-rate Grant amount of 20% of total expenditure is awarded in addition to the eligible expenditure.

4. Expected impact of project results

Please describe the expected impact of project results below (max. 600 words and up to 2 pages when including graphs).

The UHEAT project will deliver significant scientific and societal impacts by advancing our understanding of OHC changes and its role in driving MHWs, while providing actionable information for climate adaptation in fisheries management.

Scientific Impact

UHEAT will generate improved, uncertainty-aware observation-based datasets of OHC, OHT, and ocean circulation for the Atlantic and Pacific Oceans using an advanced Bayesian framework. These products will fill important spatial and temporal gaps, supporting a new level of precision in quantifying regional OHC changes and their mechanistic drivers. Additionally, through systematic analysis of observations, reanalyses, and high-resolution climate model outputs, UHEAT will deliver novel insights into the processes controlling regional OHC changes, such as the links to key ocean circulation systems (e.g., AMOC, gyre circulations) and the key pathways of heat transport. This will resolve critical knowledge gaps identified in IPCC reports, particularly concerning regional patterns of ocean warming and their drivers.

By applying emergent constraints to CMIP6 model outputs, UHEAT will reduce uncertainties in future projections of OHC changes and MHW probabilities. This approach will improve confidence in regional-scale climate projections, directly addressing IPCC assessments that currently rate confidence in regional OHC projections as low.

UHEAT will also develop an innovative AI-based prediction system for MHW occurrence that integrates OHC, OHT, circulation, and SST fields, moving beyond existing SST-only approaches. This system will support improved seasonal-to-multiyear forecasts of both surface and subsurface MHWs, offering a step-change in prediction skill relevant for marine ecosystem management.

Finally, by linking predicted and projected MHWs to shifts in fish habitat suitability, UHEAT will deliver an integrative assessment of climate-driven changes to fisheries at seasonal to long-term scales. This will inform adaptation strategies, supporting ecosystem-based fisheries management in both Europe and Japan.

UHEAT's results will be widely disseminated through publication in high-impact, peer-reviewed journals to ensure uptake by the scientific community. In addition, a project website will be developed and maintained, regularly updated by the team to share progress and scientific findings with both the research community and stakeholders.

Societal Impact

By delivering improved predictions of MHWs and their impacts on fisheries, UHEAT will provide critical information to fisheries managers, policymakers, and stakeholders seeking to adapt to ocean warming. MHWs have already caused mass mortality events, habitat shifts, and stock recruitment failures in commercial fish species. Predicting their occurrence more reliably will enable proactive management measures,

reducing economic losses and supporting food security.

By explicitly engaging with fisheries managers through planned workshops in both Japan and Europe, UHEAT will ensure that its predictive tools and ecosystem-relevant outputs directly support climate adaptation planning and sustainable fisheries management on both sides.

The project's results will also inform climate adaptation planning more broadly. Reducing uncertainty in regional OHC projections will help refine projections of regional climate change impacts, including sea-level rise and extreme events, which are partly modulated by ocean heat redistribution.

The Principal Project Leader brings experience in communicating climate science to the public, including podcast appearances for Nature and the UK National Oceanography Centre, as well as interviews with The Guardian and The Wall Street Journal. The research conducted in UHEAT will directly support and enrich these outreach activities, ensuring that the scientific advances reach wider audiences.

UHEAT's open-access data products, AI prediction tools, and habitat suitability maps will provide durable resources for the scientific community, supporting further research beyond the project's lifetime. The project will also foster stronger scientific collaboration between Europe and Japan, enhancing shared capacity to address climate adaptation challenges in the marine sector.

5. Impact on Sustainable Development Goals (SDGs)

Please describe the relevance and impact of your proposal related to the relevant SDGs. (max. 600 words and up to 2 pages when including graphs).

The UHEAT project will make targeted contributions to several United Nations Sustainable Development Goals (SDGs), supporting global efforts to address climate change, protect marine ecosystems, ensure food security, and strengthen international collaboration.

SDG 13: Climate Action

UHEAT directly supports SDG 13 by advancing our capacity to understand, predict, and adapt to climate change impacts in the ocean. MHWs are an increasingly frequent and damaging consequence of ocean warming, linked to anthropogenic greenhouse gas emissions. By improving our understanding of OHC changes and their roles as a driver of MHWs, UHEAT will reduce uncertainties in projections of regional climate impacts, including extreme ocean temperature events.

The project innovative use of emergent constraints applied to CMIP6 models will enhance the accuracy of future projections, offering more reliable information to inform national adaptation strategies against the impacts of MHWs. Furthermore, UHEAT will develop an AI-based system for seasonal-to-multiyear prediction of MHW probabilities, providing practical tools that enable proactive adaptation measures. Together, these advances will support evidence-based policy planning and strengthen resilience to climate impacts, a core target of SDG 13.

SDG 14: Life Below Water

UHEAT is highly relevant to SDG 14, which calls for the conservation and sustainable use of oceans, seas, and marine resources. MHWs pose a major threat to marine biodiversity, causing coral bleaching, seagrass loss, mass mortality events, and shifts in species distributions that undermine ecosystem structure and function.

By improving the understanding and prediction of MHW occurrence and their effects on fisheries, UHEAT will provide fisheries managers and conservation planners with actionable information to protect vulnerable marine ecosystems. The project's aim to map habitat suitability changes for commercially important fish species will enable more sustainable management of fisheries resources in both Europe and Japan,

supporting SDG 14's target to end overfishing and implement science-based management plans.

SDG 2: Zero Hunger

Fisheries and aquaculture are vital to global food security and livelihoods, particularly in coastal communities. Climate-driven changes in marine ecosystems threaten to disrupt food supply chains and reduce the availability of nutritious food. UHEAT will assess the links between MHWs and fisheries recruitment success, model future habitat suitability shifts, and deliver data products that inform adaptation planning.

By supporting more sustainable, climate-informed fisheries management, UHEAT will contribute to SDG 2's goal of ensuring access to safe, nutritious, and sufficient food year-round, especially in regions dependent on fisheries for dietary protein.

SDG 17: Partnerships for the Goals

UHEAT embodies the spirit of SDG 17 by fostering a strong international partnership between Europe and Japan. The project's joint leadership structure, with Project Leaders from Spain and Japan, ensures balanced, equitable collaboration. Regular meetings and coordinated dissemination will build mutual capacity, enhance scientific exchange, and deliver globally relevant outputs.

This transnational partnership demonstrates effective international cooperation to address global challenges, including climate change and marine ecosystem sustainability. It will also strengthen research ties between European and Japanese institutions, supporting long-term collaboration beyond the project's lifetime.

Summary

In summary, UHEAT will advance SDG 13 by supporting climate adaptation through improved understanding and prediction of MHWs. It will promote SDG 14 by enabling more sustainable management of marine ecosystems and fisheries. It will contribute to SDG 2 by supporting food security through climate-informed fisheries management. Finally, it will strengthen SDG 17 by fostering equitable, effective international collaboration.

6. Added value of multilateral cooperation

Please describe the added value of multilateral cooperation below (max. 600 words and up to 2 pages when including graphs).

UHEAT has been designed from the outset as a genuinely collaborative, transnational effort between partners in Europe and Japan. Its scientific objectives and societal impacts can only be fully achieved through this multilateral approach, which brings together complementary expertise, unique regional perspectives, and shared commitment to addressing climate adaptation challenges in the marine sector.

Complementary Expertise Across Partners

The UHEAT consortium brings together leading institutions with distinct yet highly complementary strengths. UIB (Spain) offers expertise in satellite oceanography and climate extremes, as well as leadership in Bayesian statistical methods and knowledge of AI-based prediction systems. HKU (Japan) contributes advanced experience in climate model analysis, emergent constraints, and fisheries impacts assessment. UHH (Germany) adds essential expertise in ocean circulation, including the AMOC and gyre circulations, and physical oceanographic measurements.

This unique combination of expertise enables UHEAT to address the problem of understanding OHC changes for improved MHW prediction from end to end: from improved observations, through constrained climate projections, to AI-based prediction systems and assessment of fisheries impacts. Such an integrated approach would be

challenging for any single partner to deliver alone.

Geographic and Oceanographic Scope

Multilateral cooperation is also essential given the project's geographic scope. UHEAT will analyse OHC changes and MHW occurrence in both the Atlantic and Pacific Oceans, ensuring relevance to both European and Japanese interests. The Pacific Ocean is particularly important for Japan's fisheries, while the Atlantic is critical for European climate variability and fisheries management. By combining efforts, UHEAT ensures that methodologies are harmonized, insights are transferable, and results have global relevance.

Data Sharing and Capacity Building

Multilateral cooperation will enable seamless sharing of observational datasets, model outputs, and analysis tools among partners. UHEAT will promote good practices in data and code sharing to support a culture of open science and reproducibility. There is a plan to hire early-career researchers to work on this project, who will benefit from working in an international team, building capacity on both sides of the partnership and training the next generation of climate and marine scientists in advanced observational and modelling techniques.

Improved Relevance for Fisheries Management and Policy

UHEAT's analysis of fisheries impacts will focus primarily on commercially important species in Japan, developing detailed assessments of how climate-driven habitat shifts and recruitment success may be affected by MHWs. At the same time, the project will develop a plan for extrapolating these insights and methodologies to the European context. This approach will improve the potential transferability of tools such as habitat suitability maps, supporting the design of adaptation strategies that address region-specific needs while benefiting from shared knowledge and methods. The project's outputs will therefore provide particularly valuable guidance for fisheries managers and policymakers in Japan, while laying the groundwork for future applications in Europe.

Strengthened International Collaboration and Long-Term Impact

UHEAT will establish and strengthen durable scientific links between Europe and Japan. Regular project meetings (both virtual and in-person), joint publications in peer-reviewed journals, shared data products, and coordinated dissemination strategies will build trust and ensure knowledge exchange. This will lay the foundation for continued collaboration beyond the project's lifetime, helping to address future challenges in climate adaptation, ocean observing, and fisheries management.

Summary

The added value of multilateral cooperation in UHEAT lies in the combination of complementary scientific expertise, the ability to address both Atlantic and Pacific basins in an integrated framework, enhanced data sharing and capacity building, improved policy relevance for fisheries management, and the strengthening of long-term collaborative ties between Europe and Japan. Together, these factors ensure that UHEAT's outcomes will be more impactful, globally relevant, and sustainable than would be possible through purely national efforts.

7. Short CVs of main participating researchers

Please add a short CV of the main participating researchers involved in the proposal, including a list of the five most relevant publications of the last ten years (max. 1 page per researcher).

Francisco Mir Calafat (DNI: 43118075-K)

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Affiliation: Associate Professor, Department of Physics, University of the Balearic Islands (UIB), Spain
Retained Associate of the National Oceanography Centre (NOC), UK

Education

PhD in Physics, July 2010, UIB, Spain.

Professional Experience

Associate Professor, UIB, Spain, Oct 2024 – present.

Assistant Professor, UIB, Spain, Jan 2024 – Sep 2024.

Principal Investigator, NOC, UK, 2016 – 2024.

Research Scientist, NOC, UK, 2014 – 2016.

Marie Curie Postdoctoral Researcher, University of South Florida, USA (2011–2013), and NOC, UK (2013–2014).

Key figures

- PI on 6 research projects and Co-I/Researcher on a further 18 projects.
- 50 peer-reviewed articles, including first-author papers in *Nature* and *PNAS*.
- h-index 31, citations > 3300.

Track Record

Currently, I am the PI of the CLIMEX project funded by the Spanish MICIU (CNS2023-145309) to investigate climate influences on extreme sea levels. I am also a partner on the Horizon Europe-funded EPOC project (ref: 101059547), to explain and predict the Atlantic meridional overturning circulation. I was a Co-I on EPOC while at NOC in 2022–2024. I was also the PI of an ISSI International Team in 2021–2022 (Team #524). I was multiple-PI of the 4DAtlantic-Ocean Heat Content project (2021–2023) and the Sea Level CCI project (2019–2024), both funded by the European Space Agency.

My research is organised around three core themes: the transport of heat by ocean circulation, climate influences on extreme weather, and the physics of sea-level changes. Highlights include: ♦ leading the development of a Bayesian hierarchical model for estimating Atlantic meridional heat transport from Argo, altimetry and gravimetry data [1]. ♦ Co-author of a study to estimate ocean heat content from altimetry and gravimetry data [2]. ♦ Leading the development of a Bayesian model to analyse weather extremes [3]. ♦ Leading a study showing that changes in storminess over Europe since 1960 altered the likelihood of extreme sea levels to a similar degree as sea-level rise [4]. ♦ Co-author of a study showing that the current acceleration in global mean sea level is due to increased ocean heat uptake in the Southern Ocean [5].

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Researcher, Mediterranean Institute for Advanced Studies (IMEDEA), Spain

Education

PhD in Physics, 2004, UIB, Spain

Professional Experience

Associate Professor, UIB, Spain, 2017 – present.

“Ramón y Cajal” Fellow, IMEDEA, UIB, Spain, 2009 – 2017.

Postdoctoral Researcher, National Oceanography Centre, UK, 2006 – 2008.

Research fellow, Université de La Rochelle, France, 2005 – 2006.

Key figures

- PI on 7 research projects and Co-I/Researcher on a further 13 projects.
- 145 SCI-indexed publications.
- h-index 44, citations > 6500.

Track Record

At present I am the PI of the DETECT project (2022–2025), funded by the Spanish Ministry of Science and Innovation, focused on decadal prediction of sea level extremes under climate change. I am also the UIB PI of the LIFE AdaptCalaMillor project (2023–2027), funded by the European Commission, which co-designs adaptation strategies to climate change for Mediterranean beaches. I co-led the Tech2Coast (2022–2024) and MOCCA (2019–2021) projects on coastal monitoring and morphodynamics, and I was the Spanish PI of the INSEaptION project (2017–2020), integrating sea-level projections into coastal climate services.

My research addresses key questions in coastal and climate oceanography, with emphasis on understanding sea-level variability from sub-daily to multi-decadal scales, assessing compound marine hazards, and disentangling the drivers of long-term trends and extremes. Recent contributions include the development of new methodologies to attribute marine heatwave intensification to anthropogenic warming, the characterization of storm surge–wave interactions along Mediterranean coasts, and the quantification of vertical land motion effects on global sea-level estimates. I actively contribute to the scientific community through editorial responsibilities, international panels (e.g. IPCC WGI, EuroGOOS, UNESCO/IOC), and leadership roles in European climate research initiatives.

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Retained Associate of the National Oceanography Centre (NOC), UK

Education

PhD in Oceanography, August 2009, University of Washington, USA

Professional Experience

Professor, UHH, Germany, July 2022 – present.

Science Leader, National Oceanography Centre (NOC), UK, 2022

Principal Research Scientist, NOC, UK, 2018-2022

Associate Professor, University of Southampton (UoS), UK, 2016-2018

Visiting scientist, NASA Jet Propulsion Laboratory, USA, March – June 2016

Lecturer, UoS, UK, 2012-2016

Senior Research Fellow, NOC, UK, 2009-2012

Key figures

- PI on 6 research projects and Co-I on a further 12 projects.
- 65 peer-reviewed articles, including 12 first-author papers.
- h-index 35, citations > 5200.

Track Record

Currently, I am the PI of the EPOC project funded by Horizon Europe (ref: 101059547), to explain and predict the Atlantic meridional overturning circulation (MOC). I was also PI of the RAPID project (2020—2022) to measure the Atlantic MOC at 26°N, and did a sabbatical at the NASA Jet Propulsion Laboratory to use satellite data (altimetry and gravimetry) to investigate AMOC variability.

My research is organised around two areas: the role of ocean circulation in climate and on developing methods for ocean observing. Highlights of my recent scientific contributions include: ♦ synthesizing the approaches from array-based observing of the AMOC [1,3] ♦ proof-of-concepts for using satellite data to estimate the overturning strength [2, 5] ♦ using GRACE satellite data to reduce drift in *in situ* measurements of ocean bottom pressure, towards improving estimation of overturning variability [4]. Additionally, through the EPOC project, we are using new ocean bottom pressure observations which are evaluated against satellite-based estimates of ocean pressure to reconcile identified differences in overturning calculations between arrays [1, 3].

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CURRICULUM VITAE

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Position: Associate Professor

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My research specializes in physical oceanography and fisheries oceanography, with a focus on the ocean environment and marine ecosystems from lower to higher trophic levels, across multiple spatial scales—from continental shelf to mesoscale to basin-scale—in the northwestern Pacific Ocean. My methodological strengths lie in the integration of fundamental oceanographic approaches, including:

- (A) the development of ocean state forecasting systems around Japan and the implementation of numerical simulations using coupled physical–lower-trophic-level ecosystem models [1];
- (B) field observations conducted aboard research vessels to examine environmental conditions ranging from ocean physical structures to fish feeding environments [2]; and
- (C) comprehensive analyses of in-situ observations, satellite-derived data, and model-based outputs [3].

In recognition of my contributions to the advancement of fisheries oceanography, I was awarded the prestigious Uda Prize by the Japanese Society of Fisheries Oceanography in March 2020 [4].

Recent research highlights include pioneering studies on the increasing frequency of marine heatwaves in the northwestern Pacific, investigations into the decadal-scale weakening of the western subarctic gyre and the Oyashio Current in the mid-2010s [5], and research on the offshore shift of early-season fishing grounds for Pacific saury off eastern Hokkaido [2]. These studies constitute the scientific foundation for the present proposal.

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CURRICULUM VITAE (Yoshi N. SASAKI)

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Education: Ph.D. (Science) in March, 2008 from Hokkaido University
Occupation: Associate Professor, Hokkaido University, Japan, February 2018-present
Lecturer, Hokkaido University, Japan, April 2013-January 2018
Web site: https://geodynamics.sci.hokudai.ac.jp/poc/sasakiyo/index_en.html

I study oceanic variability and the associated climate and environmental changes using various approaches, that is, theory, numerical simulation, and statistical analyses. I particularly examine ocean circulation variability in the mid-latitude [2] and its influence on the atmosphere [1]. Recently, I focus on understanding the trend components of ocean circulation, such as the warming in the East China Sea [3], and the increasing trend of the transport of the Tsushima warm current [4]. Also, I have been particularly interested in the mechanism of fluctuations of western boundary currents [5], such as the Kuroshio Extension and the Gulf Stream. I received the Okada Prize in 2015 awarded by the Oceanographic Society of Japan. I was recognized "Theoretical and analytical study for western boundary currents and their extensions".

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CURRICULUM VITAE (Shoshiro MINOBE)

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Dr. Minobe is a collaborator on this proposal. Although he cannot serve as a co-PI due to his mandatory retirement in March 2028, he is fully committed to contributing actively throughout the duration of the project.

His research spans a broad range of disciplines, including the atmosphere and physical, biogeochemical, and biological oceanography. Particularly relevant to this proposal are his findings that the dominant mode of ecosystem variability in the North Pacific over the past ~40 years is more strongly linked to global warming than to the previously assumed Pacific Decadal Oscillation [1]; that ocean deoxygenation—the loss of oxygen from the ocean—is rapidly progressing in the North Pacific, while current climate models fail to adequately represent this change [2]; that the extreme global warming observed in 2023–2024 was driven by a combination of El Niño and a marked increase in the Earth’s energy imbalance—the difference between the energy received from the Sun and that radiated back to space [3]; and that ocean heat content in the North Pacific has increased rapidly during the 2020s at the fastest rate per unit area among global oceans [4]. Notably, Dr. Minobe has analyzed more than 200 ensemble members from climate model simulations—the largest number ever used for biogeochemical climate model analysis at that time—demonstrating his strong capability in handling large-scale climate model datasets.

Dr. Minobe also has a strong international presence. He is currently a member of the World Climate Research Programme (WCRP) Lighthouse Activity, Explaining and Predicting Earth System Change; a member of the Scientific Steering Group of CLIVAR (Climate and Ocean: Variability, Predictability and Change), a core project of WCRP; and a member of the working group on Climate Extremes and Coastal Impacts in the Pacific under the North Pacific Marine Science Organization (PICES), an intergovernmental organization involving countries around the North Pacific.

His scientific achievements have been highly recognized. He received the Japan Oceanographic Society Award (March 2013), the Commendation for Science and Technology by the Minister of Education, Culture, Sports, Science and Technology (2014), and the Ninth National Maritime Award (Prime Minister’s Award) (2016). He was selected as one of the world’s 1,000 most influential climate scientists by Reuters (2021).

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8. Confirmation letter

This confirmation letter is necessary only in case of additional partners that need to secure their own funding.

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